ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

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CO2-9

EXPERIMENT 9

AIM:
To implement magnetic-field boundary conditions between two media and determine the relative permeability of the unknown medium.

Test case (i) Magnetic Boundary Conditions and Determination of the Relative Permeability

Program:

```
clc;
clear;
close all;
%% Magnetic-Field Boundary Conditions between Two Media mu0 = 4*pi*1e-7;
% H/m % Input data mur1 = input('Enter relative permeability of medium 1: ');
H1t = input('Enter tangential H-field H1t (A/m): ');
B1n = input('Enter normal flux density B1n (T): ');
H2n_meas = input('Enter measured/assigned normal H-field H2n (A/m): ');
mur2_ref = input('Enter reference relative permeability of medium 2: ');
% Permeability of the two media mu1 = mu0*mur1;
mu2_ref = mu0*mur2_ref;
% Boundary conditions for zero surface current H2t = H1t;
B2n = B1n;
% Determine relative permeability from measured H2n mur2_calc = B2n/(mu0*H2n_meas);
% Theoretical normal H-field using the reference permeability H2n_theory = B2n/mu2_ref; %
Percentage deviation dev = abs(mur2_calc-mur2_ref)/abs(mur2_ref)*100;
fprintf('\n--- Magnetic Boundary Condition Results ---\n');
fprintf('H2t (theoretical) = %.6f A/m\n',H2t);
fprintf('B2n (theoretical) = %.6e T\n',B2n);
fprintf('H2n (theoretical) = %.6f A/m\n',H2n_theory);
fprintf('Calculated mur2 = %.6f\n',mur2_calc);
fprintf('Reference mur2 = %.6f\n',mur2_ref);
fprintf('Percentage deviation = %.4f %%\n',dev);
```

% Parameter sweep: effect of relative permeability on normal H-field mur = linspace(1,10,100);

Hn = B1n./(mu0*mur);

figure;

plot(mur,Hn,'LineWidth',2);

grid on;

xlabel('Relative permeability, \mu_r');

ylabel('Normal magnetic field H_n (A/m)');

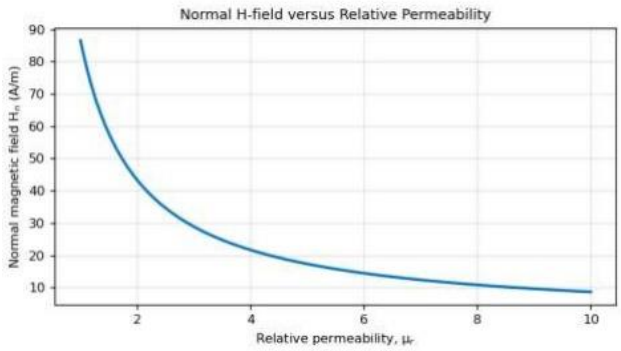
title('Normal H-field versus Relative Permeability');

% Verification of tangential H continuity and normal B continuity fprintf('\nBoundary verification for zero surface current:\n');

fprintf('H1t = %.6f A/m, H2t = %.6f A/m\n',H1t,H2t);

fprintf('B1n = %.6e T, B2n = %.6e T\n',B1n,B2n);

OUTPUT:



Test case (ii) Field Components and Refraction Angle at the Interface of Two Magnetic Media

program:

clc;

clear;

mu0 = 4*pi*1e-7;

H1 = input('Enter magnitude of H1 in medium 1 (A/m): ');

th1 = input('Enter angle of H1 with the normal (deg): ');

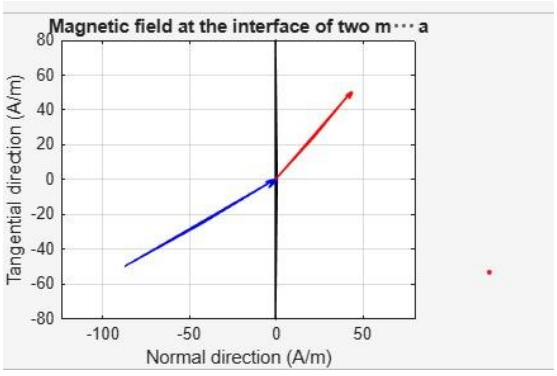
mur1 = input('Enter relative permeability of medium 1: ');

mur2 = input('Enter relative permeability of medium 2: ');

H1n = H1*cosd(th1);

```
 % normal component in medium 1 H1t = H1*sind(th1);
 % tangential component in medium 1 B1n = mu0*mur1*H1n;
% normal flux density in medium 1 H2t = H1t;
 % tangential H is continuous (Ks = 0) B2n = B1n;
% normal B is continuous H2n = B2n/(mu0*mur2);
 % normal H in medium 2 H2 = sqrt(H2n^2 + H2t^2);
 th2 = atand(H2t/H2n);
fprintf('H1n = %.2f A/m, H1t = %.2f A/m\n', H1n, H1t);
fprintf('B1n = %.4e T\n', B1n);
 fprintf('H2t = %.2f A/m, B2n = %.4e T\n', H2t, B2n);
 fprintf('H2n = %.2f A/m, |H2| = %.2f A/m\n', H2n, H2);
 fprintf('Refraction angle theta2 = %.2f deg\n', th2);
figure;
 plot([0 0],[-80 80],'k','LineWidth',2);
hold on;
 quiver(-H1n,-H1t,H1n,H1t,0,'b','LineWidth',2);
 quiver(0,0,H2n,H2t,0,'r','LineWidth',2);
 axis equal;
 grid on;
xlabel('Normal direction (A/m)');
 ylabel('Tangential direction (A/m)');
 title('Magnetic field at the interface of two media');
```

OUTPUT:

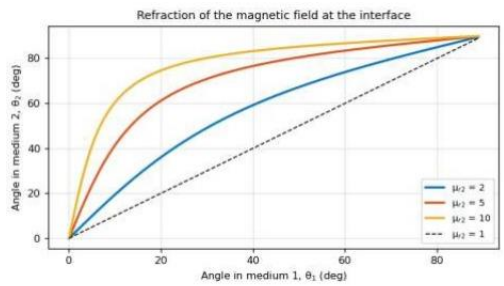


Test case (iii) Variation of the Refraction Angle with the Angle of Incidence for Different Permeabilities:

Program:

```
clc;
 clear;
 th1 = linspace(0,89,179);
mur1 = 1; mur2 = [2 5 10];
figure;
hold on;
for k = 1:3 th2 = atand((mur2(k)/mur1)*tand(th1));
  % tan(th2) = (mur2/mur1)*tan(th1) plot(th1,th2,'LineWidth',2);
 end plot(th1,th1,'k--');
% mur2 = mur1 (no change of direction) grid on;
xlabel('Angle in medium 1, \theta_1 (deg)');
ylabel('Angle in medium 2, \theta_2 (deg)');
 legend('mur2 = 2','mur2 = 5','mur2 = 10','mur2 = 1','Location','southeast');
 title('Refraction of the magnetic field at the interface');
```

OUTPUT:

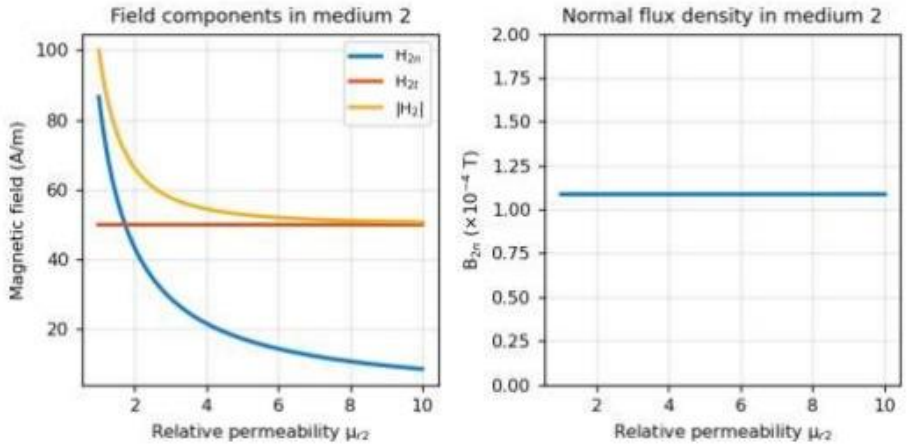


Test case (iv) Effect of the Relative Permeability on the Field Components in Medium 2:

Program:

```
clc;
clear;
mu0 = 4*pi*1e-7; H1 = 100;
 th1 = 30;
mur1 = 1;
H1n = H1*cosd(th1);
H1t = H1*sind(th1);
```

```
B1n = mu0*mur1*H1n;
 mur2 = linspace(1,10,100);
 H2n = B1n./(mu0*mur2);
 % normal H in medium 2 H2t = H1t*ones(size(mur2));
% tangential H (continuous) H2 = sqrt(H2n.^2 + H2t.^2);
B2n = mu0*mur2.*H2n; % normal B (continuous) subplot(1,2,1);
plot(mur2,H2n,mur2,H2t,mur2,H2,'LineWidth',2);
grid on; xlabel('Relative permeability mur2');
 ylabel('Magnetic field (A/m)');
 legend('H2n','H2t','|H2|'); subplot(1,2,2);
plot(mur2,B2n,'LineWidth',2);
 grid on;
 ylim([0 2e-4]);
 xlabel('Relative permeability mur2');
ylabel('B2n (T)');
```



Result :

Thus the Implementation of Magnetic-Field Boundary Conditions between Two Dielectric Media and Determination of Relative Permeability using MATLAB successfully.

LABORATORY PERFORMANCE ASSESSMENT RUBRICS

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weighting**	**Marks given**	**Marks obtained**
Identify magnetic media and boundary orientation	Identify magnetic media and boundary orientation	Correctly identifies isolatedsphere radius or inner/outer radii and medium permittivity	Setup is correct with minor omissions.	Needs help identifying components or assumptions.	Uses incorrect interface definitions	25	2.5	
Apply magnetic boundary relations.	Apply magnetic boundary relations	Correctly applies continuity of normal B and the tangential H relation.	Relations are applied with minor errors.	Partly correct application requires guidance.	Boundary relations are incorrect.	25	2.5	
Determine permeability and validate fields	Determine permeabilit y and validate fields	Calculates required quantities and verifies the boundary relations consistently.	Results are mostly correct with minor discrepancies.	Validation is incomplete.	Results are invalid or unverified.	30	3	
Explain and document findings	Explain and document findings	Explains magnetic-field behaviour and records assumptions, units and conclusion.	Documentation is clear with minor omissions.	Interpretation or documentatio n is incomplete.	Cannot explain or report results.	20	2	
TOTAL						100	10	